CYBERSECURITY PROJECT

TOPIC OUTLINE & WORK ASSIGNMENTS

TOPIC 1: COMMON THREATS

Application Vulnerabilities

1. SQL Injection (SQLi) – Manipulation of the Query Abstract Syntax Tree (AST). **[Adam]**

2. Cross-Site Scripting (XSS) – Stored vs. Reflected XSS and DOM Manipulation. **[Adam]**

3. Broken Access Control and IDOR (Insecure Direct Object References). **[Adam]**

Network Threats

4. Man-in-the-Middle (MitM) Attacks and ARP Cache Poisoning – OSI Layer 2. **[Adam]**

Availability Threats

5. DDoS (Distributed Denial of Service) Attacks – Volumetric, Protocol-based and Layer 7 attacks. **[Jiaming]**

Authentication Attacks

6. Authentication Attacks – Credential Stuffing, Brute Force and Password Spraying. **[Jiaming]**

7. Technical Phishing, Spear Phishing and Email Spoofing Infrastructure. **[Jiaming]**

Advanced Threats and Supply Chain Attacks

8. Software Supply Chain Attacks and Dependency Confusion. **[Jiaming]**

TOPIC 2: MALWARE

Malware Taxonomy

9. Computer Viruses – Binary File Infection Techniques (PE / ELF). **[Marco]**

10. Worms – Autonomous Network Propagation and RCE Exploitation. **[Marco]**

11. Trojan Horses, Droppers and RATs (Remote Access Trojans). **[Marco]**

12. Ransomware – Hybrid Cryptographic Architecture (Crypto-Locking Model). **[Marco]**

13. Spyware, Keyloggers and Infostealers – Input and Credential Capture Mechanisms. **[Lazar]**

Control Infrastructure

14. Botnets and Command and Control Architectures (C2 / C&C). **[Camilla]**

System Persistence

15. Operating System Persistence Mechanisms and Anti-Analysis Techniques. **[Camilla]**

Concealment and Evasion

16. Rootkits – User Mode (Ring 3) vs. Kernel Mode (Ring 0) and API Hooking. **[Camilla]**

17. Evasion Techniques – Packers, Crypters, Polymorphism and Metamorphism. **[Camilla]**

WORK DISTRIBUTION

Adam: Topics 1–4 **Jiaming:** Topics 5–8
Marco: Topics 9–12 **Lazar:** Topic 13
Camilla: Topics 14–17